

and development of enterprise mobility solutions, mobile device management solutions and Android-based custom device solutions for enterprises. She is currently involved in assessing the challenges and opportunities of embedded Android for enterprises in various verticals. She says, "I started working on Android app development ever since it made an entry into the smartphone market. Android's market share and feature-richness makes it a versatile platform empowering the application developer to achieve complex application use cases for a rich user experience. Android's business-friendly licence, mature application ecosystem, customisability and hassle-free supply chain commoditisation has led to a huge market of Android-powered smartphones, tablets and more. Clearly, Android is no more only a common man's companion on a smartphone or tablet, but is also used in automobiles as in-vehicle infotainment systems (IVI), field force terminals, medical devices, retail kiosks, patient monitoring terminals, etc. Considering this, it is inevitable for any person (including me) working in the consumer or enterprise mobility domains to be familiar with Android app development."

What hiring managers look for

At a time when the job trends in the Android market look upbeat, what are HR managers looking for in prospective candidates? Sachin Naik, who affirms that Xolo hires Android developers at the entry and senior levels, says, "The core strength for any developer needs to be programming. So candidates will be evaluated for knowledge of the Android SDK, their applications' life cycle, object oriented programming and core Java. Though the Android application environment abstracts out the intricacies of the chip set, network and power management, the developers need to be aware of them. This ensures that the applications they create do not consume power or network bandwidth unnecessarily. It's a plus if candidates have a basic level of understanding about the underlying system and hardware. Android application development also involves specialisation in 2D and 3D graphics design for gaming, and NDK-based development for hardware. Sometimes, we do hire for specific areas."

Get the right skill sets

Industry pundits say it is important to acquire the right skill sets and attitude to secure a job in the Android app development market. So, what are the skills one must have to stay ahead of the curve? "From a coding perspective, the No 1 skill needed is an understanding of the OS and how apps run on it. This allows developers to write efficient code that uses less resources and works better with other apps in the ecosystem. Core Java and an understanding of Android development tools are essential to write better applications. Like any other mobile OS, memory is managed differently when compared to servers, which most people have an understanding of. It is important to accommodate that difference in the way you write code. Mobile developers have to work with the UX/UI of the application. For this, an Android app developer needs to be aware of the

Things to be kept in mind while developing an Android app

- **Resolution-neutral UI:** The application must take advantage of Android's modular structure to specify layouts and resources specific to the target form factors and DPI range. Additionally, all the relative layout calculations should be done with respect to the screen width and height (fetched from the official source giving device information) and dimensions. xml should be leveraged for specifying the same.
- **Separation of controller and business logic:** As we know, the *Activity* derived classes defined in the app, control the various views of the app. And the activity life cycle is not under the control of the app itself. Hence, the complete business logic including all data structures must reside in a separate class other than *Activity*, so that the business logic data structures do not get re-initialised when the activity gets re-created.
- **Low memory handling:** The app must seamlessly handle this condition and release the memory savvy resources at the earliest, prepare to save the state of the app and exit, if required.
- **Low battery handling:** The app must seamlessly handle this condition and release the battery savvy resources at the earliest, prepare to save the state of the app and exit if required.
- **Caching:** Input/output operations incur significant costs in mobile apps in terms of battery usage, memory, data network bandwidth and latency time. Hence, it would be wise to cache the data structures frequently required by the app.
- **Multi-threading:** While complex apps do need to spawn multiple threads for background processing, achieving this in Android by way of services is pretty straightforward and is the preferred way.
- **Broadcasting intents:** While broadcasting intents, it is advisable to direct the broadcast for the specific target class only. Otherwise, the IPC will be sensitive to any other hacker app installed on the device.
- **Background operations:** I/O, databases, networks and other time consuming operations must be done as part of *AsyncTask* or a separate service. Also, the application must show some progress indication while any operation is in progress for over a few seconds.
- **Inter-process communication (IPC):** When large data structure transfers are required through IPC, use Android Interface Definition Language (AIDL) instead of intent-based IPC. Android has superior IPC support in AIDL to deal with such cases efficiently.

design guidelines, coding best practices and user interaction patterns," says Amrit.

Challenges faced while developing a mobile app

When we asked the developers about the major challenges they stumble upon while developing an app, they came up with a long list. According to Pooja Maheshwari, some of the major challenges are: